LexiPic

Language and Heritage Archive

Complete Suite Documentation

LexiPic Plugin · LexiPic Kiosk Plugin · Kiosk PWA

Version 1.2.x
BhoomiTech Heritage and Development Foundation

1. Suite Overview

LexiPic Language and Heritage Archive is an integrated, open-source suite of three components designed to document, distribute, and display heritage-language word sets across any combination of WordPress websites, museum kiosks, classroom touchscreens, and mobile devices.

1.1 What the Suite Does

Each entry in a LexiPic archive pairs a photograph of an object or concept with its spoken pronunciation, its written form in native script, and a Roman transliteration — building a structured, multimedia "picture dictionary" for an endangered or under-resourced language. Entries are grouped into named Sets (e.g. "Domestic Animals", "Kitchen Items", "Numbers 1–10") and delivered as interactive swipeable card sliders.

1.2 Component Map

Component	Role
LexiPic Plugin (v1.2.2)	Author and archive word sets inside any WordPress site. Provides the content-creation workspace, the IME transliteration engine, REST API, and JSON import/export.
LexiPic Kiosk Plugin (v1.2.1)	Distribute and display sets on dedicated kiosk screens from a WordPress site. Provides set management, a fullscreen kiosk URL, shortcode embedding, PIN-protected admin panel, and the domain allowlist API.
Kiosk PWA (v2.1.0)	The front-end player that runs on kiosk devices, PCs, or mobiles. Connects to a remote WordPress Kiosk site via URL, or plays data loaded from a USB drive — fully offline-capable.

1.3 How the Components Work Together

The typical workflow moves content from left to right through three stages:

- Create — A contributor logs into a WordPress site running the LexiPic Plugin, opens the [lexipic] shortcode page, and uses the three-tab workspace to photograph objects, record pronunciations via the browser microphone, and type words using the built-in phonetic IME. Entries are saved into the site's database and can be exported at any time as a self-contained JSON file.
- Distribute — The site operator installs the LexiPic Kiosk Plugin on the same (or a different) WordPress site, imports the JSON export, and either embeds the kiosk via shortcode or uses the dedicated fullscreen URL. The Kiosk Plugin's REST API serves set data to any connected device.

- Display — The Kiosk PWA runs on a kiosk touchscreen, a library PC, or a visitor's mobile. In Remote URL mode it points at the Kiosk Plugin's fullscreen page and renders it inside a sandboxed full-screen wrapper with automatic reload recovery. In USB mode it plays JSON data directly from a folder on a USB drive — no internet required.

> *All three components are released under the GNU General Public License v2 or later (GPLv2+). The full suite is developed and maintained by the BhoomiTech Heritage and Development Foundation.*

2. LexiPic Plugin

Plugin slug: lexipic-plugin **Version:** 1.2.2 **Requires:** WordPress 5.8+, PHP 7.4+

The LexiPic Plugin turns any WordPress page into a structured multimedia archive for a heritage language. It is the authoring component of the suite — the place where contributors capture photographs, record speech, and type words in native script.

2.1 Installation

1. Upload the lexipic-plugin folder to /wp-content/plugins/, or install the .zip via Plugins → Add New → Upload Plugin.
2. Activate the plugin from the WordPress Plugins menu.
3. Navigate to LexiPic → Settings to choose the default language, configure the public archive toggle, and set per-tab access control.
4. Add the shortcode [lexipic] to any page or post. Use [lexipic set="your-set-slug"] to pre-load a specific set on load.
5. (Optional) Visit LexiPic → IME Engine to register additional languages or upload custom transliteration map files.

2.2 The Three-Tab Front-End Workspace

Placing [lexipic] on a WordPress page renders a self-contained three-tab application. Tab visibility is controlled by the administrator; visitors who lack permission see only the Archive tab.

Archive Tab

Displays all published entries in the currently selected set as a swipeable card slider. Each card shows the photograph, the word in native script, the Roman transliteration, an optional description, and a Play button for the voice recording. Visitors can switch between sets using a set selector.

Add Entry Tab

A contribution form that captures all four data layers for a new word:

- Photo — taken with the device camera or uploaded from disk.
- Voice Recording — captured directly in the browser using the Web Speech API microphone "Speak" control. On browsers without Web Speech API support, the pronunciation field can be filled manually.
- Native Script — typed using the built-in phonetic IME. As the contributor types Roman characters, the IME engine converts them in real time to the native script.

- Roman Transliteration — generated automatically from the typed native script via the reverse map, or typed directly.
- Description / Note — an optional free-text field for cultural context, usage notes, or a sentence example.

Sets Tab

Lets authorised users create new sets, rename or delete existing ones, and trigger JSON import/export. Exports include all entries, their photographs (base64-encoded), and their audio recordings.

2.3 The CompLing AI IME Engine

The built-in phonetic Input Method Engine is a deterministic, rule-based transliteration system — never AI-generated at runtime. It reads per-language JSON configuration files:

File	Purpose
forwardMap.json	Maps Roman key sequences to native script characters (e.g. "k" → "ক").
reverseMap.json	Maps native script characters back to Roman (for auto-filling the transliteration field from typed script).
defaults.json	The complete character table for a language, used to populate the on-screen character picker.
js-engine-settings.json	Controls virama, ZWJ/ZWNJ insertion, inherent vowel, linking behaviour, and section display order.
engine.js	The core transliteration logic. Shared across all languages; language behaviour is driven entirely by the JSON files.

Three languages ship out of the box: Bishnupriya Manipuri (ইমার ঠার, code bpm), Assamese (অসমীয়া, code asm), and Bengali (বাংলা, code ben). Each has its own speech-recognition locale.

Adding a New Language

6. Go to LexiPic → IME Engine and click Add Language.
7. Enter a unique language code (e.g. mni) and a human-readable label.
8. Optionally clone an existing language's engine files as a starting point.
9. Upload your own forwardMap.json, reverseMap.json, defaults.json, and js-engine-settings.json files.

10. The plugin validates and stores the files under wp-content/uploads/lexipic/ime/{language-code}/. The last 20 versions of each file are automatically backed up and can be restored from the IME Engine page.

2.4 Access Control

Administrators always have full access. All other users are governed by the Tab Access Control setting:

- Public Archive — toggle whether logged-out visitors can view the Archive tab. Off by default.
- Add Entry tab — restricted to a comma-separated list of WordPress usernames (plus all admins).
- Sets tab — same; can be a different list from the Add Entry tab.
- Anyone not on the relevant list sees only the Archive tab, regardless of their WordPress role.

2.5 Audio Handling and FFmpeg

Recordings captured in the browser are saved as .webm files under wp-content/uploads/lexipic/. If FFmpeg is installed on the server, the plugin automatically converts new recordings to .mp3 in the background a few seconds after upload, improving playback compatibility on Safari and iOS. The plugin detects FFmpeg automatically; the Settings page shows its detection status. If FFmpeg is absent, recordings remain as .webm — playback still works in most modern browsers.

2.6 JSON Import and Export

Each set can be exported as a single, self-contained JSON file that includes all text fields and base64-encoded copies of every photograph and audio recording. The file can be:

- Re-imported into the same site (for backup/restore).
- Imported into a different WordPress site running the LexiPic Plugin.
- Imported into the LexiPic Kiosk Plugin for kiosk display.
- Placed in the lexipic-kiosk-data/ folder on a USB drive for offline Kiosk PWA playback.

Legacy exports from the original LexiPic PWA prototype (pre-1.0) are also accepted on import.

2.7 REST API (lexipic/v1)

All front-end interactions go through the plugin's REST API — the WordPress database is never accessed directly from JavaScript. The full namespace is lexipic/v1.

Method	Endpoint	Description

GET	/sets	List all sets.
POST	/sets	Create a set (name, language).
GET	/sets/{id}	Get one set (metadata only).
DELETE	/sets/{id}	Delete a set and all its entries and files.
GET	/sets/{set_id}/entries	List entries for a set (no image blobs).
POST	/sets/{set_id}/entries	Create an entry (image, audio, script, roman, description).
DELETE	/entries/{id}	Delete a single entry.
GET	/entries/{id}/image	Serve the entry's image blob (byte-range supported).
GET	/sets/{id}/export	Download a full JSON export for a set.
POST	/import	Import a JSON export file.

2.8 Data Storage

Data Type	Storage Location
Entry text (script, roman, description)	WordPress database table: wp_lexipic_entries
Entry photographs	Binary stored in the database (image_data column)
Audio recordings	Files under wp-content/uploads/lexipic/
IME engine overrides	Files under wp-content/uploads/lexipic/ime/{lang-code}/
Plugin settings	WordPress options table (lexipic_settings)

2.9 Uninstallation

Deactivating the plugin does not remove any data. Deleting the plugin from the Plugins → Installed Plugins screen permanently removes its database tables (wp_lexipic_sets and wp_lexipic_entries), all uploaded audio files, all custom IME engine files, all plugin options, and all scheduled background events. Export any sets you want to keep before deleting.

3. LexiPic Kiosk Plugin

Plugin slug: lexipic-kiosk **Version:** 1.2.1 **Requires:** WordPress 6.0+, PHP 7.4+

The LexiPic Kiosk Plugin is an independent WordPress plugin that acts as the distribution and display server for the suite. It receives word-set JSON exports (from the LexiPic Plugin or created manually), manages them in its own database, and serves them to kiosk screens — either via a shortcode embedded in a themed WordPress page, via a dedicated fullscreen URL, or remotely through its REST API to the Kiosk PWA.

3.1 Installation

11. Upload the lexipic-kiosk folder to /wp-content/plugins/, or install the .zip via Plugins → Add New → Upload Plugin.
12. Activate the plugin from the WordPress Plugins menu.
13. Go to LexiPic Kiosk → Settings and set a kiosk admin PIN. The on-screen admin panel remains locked until a PIN is configured.
14. Go to LexiPic Kiosk → Word Sets and import your first .json export.
15. Either add [lexipic_kiosk] to any page, or use the dedicated kiosk URL shown under LexiPic Kiosk → Embed & Kiosk URL.

3.2 Admin Panel (wp-admin)

The plugin adds a LexiPic Kiosk top-level menu to wp-admin with the following sections:

Word Sets

Import .json exports, rename sets, change their display order (drag-and-drop), toggle sets active or inactive, and delete unwanted sets. An inactive set is hidden from the kiosk display and from the Kiosk PWA but remains in the database. The page also provides a full Backup (export all sets to one JSON file) and Restore (replace all sets from a backup file) function.

Settings

Configure the kiosk admin PIN, the default idle-timer duration (how many seconds of inactivity before the kiosk returns to the set-selection grid), and the default autoplay-audio behaviour. These settings are site-wide defaults; kiosk operators can adjust the idle timer per device from the on-screen panel. The Settings page also includes a "Delete all data on uninstall" checkbox — data is preserved by default when the plugin is deleted unless this is checked first.

Embed & Kiosk URL

Shows the exact shortcode ([lexipic_kiosk]) and the dedicated fullscreen URL (either /?lexipic_kiosk=1 or /kiosk/ depending on permalink settings). Copy either to deploy the kiosk.

Approved Kiosk URLs (Allowlist)

A curated list of hostnames that the standalone Kiosk PWA is permitted to connect to in its Remote URL mode. Only exact hostname matches are accepted — no wildcards. This page has no effect on the plugin's own embedded kiosk. See Section 4 (Kiosk PWA) and Section 5.3 (Allowlist Security) for the full security model.

3.3 The On-Screen Admin Panel (PIN-Protected)

A kiosk operator who knows the PIN can manage the kiosk without a WordPress login. Accessed by tapping a small gear icon on the kiosk interface and entering the PIN, the panel provides:

- Import a new set from a JSON file (via file picker on the kiosk device).
- Rename sets.
- Reorder sets.
- Toggle sets active or inactive.
- Delete sets.
- Backup (download all sets as JSON) and Restore.
- Adjust the idle-timer for this device (local only — does not change the site-wide default).

3.4 Kiosk Display Modes

Shortcode Embed ([lexipic_kiosk])

Renders the kiosk inside a normal WordPress page, within the active theme's header and footer. Suitable for a library page or a classroom portal where the kiosk is one element among others.

Dedicated Fullscreen URL

A theme-agnostic URL that renders only the kiosk — no WordPress header, footer, admin toolbar, or theme CSS. Optimised for dedicated hardware running in a locked-down browser. The URL is: https://yoursite.com/?lexipic_kiosk=1 (or https://yoursite.com/kiosk/ with pretty permalinks enabled). The admin toolbar is suppressed even for logged-in administrators when this URL is accessed directly.

3.5 Offline Resilience

The kiosk caches its set data in the browser's IndexedDB storage. A brief network interruption does not interrupt a visitor's session — the kiosk continues to display from its local cache. WordPress remains the single source of truth; the cache is refreshed each time the kiosk loads or re-syncs.

3.6 Media Storage

On import, the plugin converts all base64-encoded image and audio data from the JSON file into ordinary files stored under wp-content/uploads/lexipic-kiosk/. Files are referenced by URL in the database. They are not registered in the WordPress Media Library. Exports and backups re-encode the files as base64 to produce a single portable JSON file. Importing from older backups that already contain base64 media works without any conversion step.

3.7 REST API (lexipic-kiosk/v1)

The Kiosk Plugin exposes a REST API used by both its own front-end JavaScript and the standalone Kiosk PWA. "Protected" endpoints accept either a logged-in WordPress user with the manage_lexipic_kiosk capability, or a valid kiosk session token obtained from /pin/verify via the X-Lexipic-Kiosk-Token header.

Method + Endpoint	Auth	Description
GET /sets	Public	List all sets (metadata). Pass ?include_entries=1 for full entries.
GET /sets/{slug}	Public	Single set with all entries.
POST /sets	Protected	Create or overwrite a set (import).
POST /sets/{slug}/rename	Protected	Rename a set.
POST /sets/{slug}/active	Protected	Toggle a set's active flag.
POST /sets/reorder	Protected	Swap two sets' display order.
DELETE /sets/{slug}	Protected	Delete a set.
GET /settings	Public	Read idle timer and autoplay defaults (no PIN hash).
POST /settings	Admin only	Update site-wide idle timer and autoplay defaults.
POST /pin/verify	Public	Exchange a PIN for a session token.
GET /allowlist	Public	Approved domain list for Kiosk PWA Remote URL mode.
POST /allowlist	Admin only	Replace the approved domain list.
GET /backup	Protected	Download full export payload.
POST /restore	Protected	Destructive full restore from a backup file.

POST /flush	Protected	Wipe all sets and data.

4. Kiosk PWA (kiosk-card-nt)

Package: lexipic-pwa-nt **Version:** 2.1.0 **Type:** Progressive Web App (no server required)

The Kiosk PWA is a fully self-contained Progressive Web App — a single folder of HTML, CSS, and JavaScript with no server-side component. It runs in any modern browser and can be installed as a standalone fullscreen app on a kiosk device, a desktop PC, or a mobile phone. It is the display engine for heritage word sets, and it supports two entirely different source modes.

4.1 Deployment

16. Copy the lexipic-pwa-nt folder to any web server, USB drive root, or local filesystem accessible by the kiosk device's browser.
17. Open index.html in the browser. On first launch, the Setup screen appears.
18. Choose a source mode (see below) and configure it. The choice is saved and the kiosk restores it automatically on every subsequent load.
19. (Optional) Install the PWA from the browser's "Install app" prompt to run it as a standalone fullscreen application with its own icon, no browser chrome, and automatic offline caching via the service worker.

4.2 PWA Variants

The Kiosk PWA is distributed in two variants. Both share the same folder structure, set-data format, and display interface; they differ only in how they handle PIN verification when the kiosk has never been online.

Variant	Package / Use case
kiosk-card-nt (Network-required)	lexipic-pwa-nt — for kiosks that have internet access at least occasionally. PIN verification always requires a live network round-trip. Described throughout this section.
kiosk-card-sa (Standalone / offline)	lexipic-pwa — for kiosks that have NO internet access at all. Includes a one-time PIN bypass for first-time setup only. See the box below.

kiosk-card-sa — Standalone (Offline) PWA Variant

Use this variant (lexipic-pwa) when the kiosk device has no internet connection at any point.

The standard PIN-verification mechanism requires a live network round-trip to the WordPress Kiosk Plugin's /pin/verify endpoint every time a source change is made. On a kiosk that is permanently offline this creates an inescapable deadlock:

- First-time setup succeeds — no PIN is required yet, because nothing has been saved to protect.
- The operator then needs to change the source — even to correct a first-day mistake — and is immediately locked out, because the PIN check can never reach the server.
- The kiosk is permanently stuck, with no recovery path short of clearing all browser storage.

How the bypass works

The SA variant tracks whether this specific device has ever successfully completed a PIN verification online (stored in localStorage under the key lexipic_pin_verified_once). Until that happens, the "Change source" screen shows a clearly-labelled bypass button alongside the normal PIN field:

```
Continue without PIN (first-time setup)
```

The bypass is intentionally limited and self-expiring:

- It appears only while the device is in the "unclaimed" state — i.e. a PIN has never been verified online on this device.
- The moment any PIN is successfully verified online, the bypass is permanently hidden on this device and every future source change requires the real, network-verified PIN — exactly as in the NT variant.
- Physical access to the device is already required to tap the bypass button — the same access that picking a USB folder requires. No new attack surface is introduced.

Which variant to choose: If the kiosk will ever connect to the internet — even briefly during initial commissioning — use `kiosk-card-nt`. Only use `kiosk-card-sa` for permanently air-gapped kiosks running entirely on USB data.

4.3 Source Modes

Remote URL Mode

The PWA displays a remote WordPress page (typically the LexiPic Kiosk Plugin's fullscreen URL) directly inside a sandboxed fullscreen iframe. The PWA itself does not parse or interpret the JSON data — it simply presents the page, exactly as a dedicated kiosk browser pointed at a URL would. Key behaviours:

- If the page fails to load (offline, DNS error, server restart), a retry screen appears automatically and the PWA keeps attempting to reload every 8 seconds until it succeeds.
- The configured URL is saved in localStorage and restored automatically on every reload — no re-entry needed after a power cycle.

- Every URL entered is validated against the domain allowlist served by the Kiosk Plugin before being saved (see Section 5.3). Changing an already-saved source additionally requires the kiosk PIN.
- The allowlist is re-checked periodically while the kiosk is running (every few minutes). A URL that is removed from the allowlist after being saved will be revoked on the next recheck, with a short countdown warning before disconnecting.

USB Mode

The PWA reads word-set data directly from a folder on a USB drive (or any local directory) using the browser's File System Access API. No network connection is required after the initial folder selection. Key behaviours:

- On first use, the user clicks "Select folder…" and grants the browser permission to read the lexipic-kiosk-data directory.
- The directory handle (not just the name) is saved in IndexedDB. On subsequent loads, the PWA silently re-opens the saved handle; if the browser has already granted permission it requires no user interaction at all.
- If permission was revoked or the handle is unreadable, the setup screen shows a "Reconnect to saved USB folder" button that requires only a single tap — the full folder picker is not shown again unless the handle is completely gone.
- Set data is rendered entirely client-side using the PWA's own tile-grid and card-slider UI.

4.4 Folder Structure (USB Mode)

The lexipic-kiosk-data folder must contain a settings.json file and one .json file per word set:

```
lexipic-kiosk-data/
  settings.json            ← required: lists which sets to show
  domestic-animals.json    ← one file per set (slug matches settings.json)
  numbers.json
  colours.json
  cat.jpg                  ← optional: image files if not base64-embedded
```

settings.json Format

```
{
  "layers": [
    { "slug": "domestic-animals", "name": "Domestic Animals",
      "enabled": true, "group": "Nature" },
    { "slug": "numbers", "name": "Numbers 1-10",
      "enabled": true, "group": "Learning" },
    { "slug": "colours", "name": "Colours",
      "enabled": false, "group": "Learning" }
```

```
  ],
  "kiosk": {
    "idle_time_seconds": 90,
    "autoplay_audio": false
  }
}
```

Field	Description
layers[].slug	Filename of the set JSON (without .json extension).
layers[].name	Display name shown on the tile grid (overrides the set's own name).
layers[].enabled	false hides the set from the tile grid without deleting it.
layers[].group	Section header label on the tile grid (e.g. "Nature", "Food Habit").
kiosk.idle_time_seconds	Seconds of inactivity before returning to the set-selection grid.
kiosk.autoplay_audio	If true, a word's audio plays automatically when its card opens.

4.5 Set JSON Format

Each {slug}.json file uses the same format as a LexiPic Plugin export. Images and audio can be data: URIs embedded directly in the JSON, or plain filenames relative to the data folder (e.g. "image": "cat.jpg").

```
{
  "lexipic_version": "1.1.0",
  "set": {
    "name": "Domestic Animals",
    "slug": "domestic-animals",
    "language": "bpm"
  },
  "entries": [
    {
      "id": 1,
      "word_script": "মেকুর",
      "word_roman": "mekur",
      "description": "A common household cat.",
      "image": "data:image/jpeg;base64,/9j/4AAQ...",
      "audio": "data:audio/webm;base64,GkXfo...",
      "image_mime": "image/jpeg",
      "audio_mime": "audio/webm"
    }
  ]
```

```
}
```

4.6 PWA Features

- Service Worker — a service worker (sw.js) caches the app shell (index.html, app.js, kiosk.css, icons, Material Icons font). The app loads instantly on repeat visits and functions offline in USB mode.
- Installable — the Web App Manifest (manifest.json) declares the app as installable. Browsers show an "Install" prompt; once installed, the PWA runs fullscreen with no browser chrome.
- Material Icons (vendored) — icon glyphs are bundled locally rather than loaded from a CDN. This ensures every icon renders correctly on the kiosk's very first launch with zero network connectivity. The decorative heading font (Fraunces) is CDN-loaded since its absence is a cosmetic-only issue with a serif fallback in place.
- Idle Reset — after the configured idle period of inactivity, the kiosk automatically returns to the set-selection grid, ready for the next visitor.

4.7 Bundled Sample Data

The PWA ships with four example sets for Bishnupriya Manipuri (bpm), pre-configured in settings.json:

Set	Group
bm-dom-animal — Domestic Animals	Livestock
bm-fish — Fishes	Food Habit
bm-fruits — Fruits	Food Habit
bm-vegetable — Vegetables	Food Habit

These sets are intended as working demonstrations. Replace or supplement them with your own exported sets for production deployment.

5. Security Architecture

The suite has several overlapping security mechanisms that are worth understanding as a whole, particularly for deployments in public-facing environments such as museums or libraries.

5.1 LexiPic Plugin Access Control

The LexiPic Plugin REST API enforces two permission levels. Read access (viewing sets and entries) is controlled by the "Public Archive" toggle — either any visitor or only logged-in users. Write access (creating sets, adding entries, importing data) is always restricted to logged-in users who appear on the relevant access-control list. Administrators always have full access. No anonymous user can ever modify content, regardless of the public archive setting.

5.2 Kiosk Plugin — Dual Authentication

Protected REST endpoints in the Kiosk Plugin accept two types of credentials:

- A logged-in WordPress session with the manage_lexipic_kiosk capability (for remote administration from wp-admin).
- A kiosk session token obtained by calling POST /pin/verify with the correct PIN (for on-device management without a WordPress login).

The PIN is stored hashed server-side; the plaintext PIN is never stored. Rate limiting is applied to /pin/verify to resist brute-force attacks. A PIN-holder can manage sets and adjust local settings but cannot change the site-wide idle-timer default or the approved-domain allowlist — those require a full WordPress admin session.

5.3 Kiosk PWA — Domain Allowlist

The domain allowlist is a key integrity control for the Kiosk PWA's Remote URL mode. Its purpose is to ensure that a kiosk device can only ever display content from sites that have been explicitly approved by a WordPress administrator — preventing a malicious actor from pointing a kiosk at an arbitrary external site by typing a URL into the setup screen.

> *The allowlist is about integrity, not secrecy. The list itself is publicly readable via the GET /allowlist endpoint. What it prevents is unauthorized sources from being saved to or persisted on a kiosk device.*

Save-time Validation (Fails Closed)

When a user enters a new URL on the PWA setup screen, the PWA fetches the allowlist from the configured ALLOWLIST_ENDPOINT and checks that the URL's hostname is an exact match (no wildcards). If the endpoint cannot be reached and no cached allowlist exists from a previous successful fetch, the save action is blocked — not allowed through unchecked.

Runtime Re-validation (Fails Open on Network Error)

While the kiosk is running, the allowlist is re-checked periodically (every few minutes). If the re-check finds that the current source URL is no longer in the allowlist, a short countdown warning is shown on screen before the kiosk disconnects and returns to the setup screen. However, if the allowlist endpoint is simply unreachable during a periodic re-check (e.g. a brief network blip), the currently-running session continues — a network interruption must not turn into a kiosk outage for a legitimately-saved URL. Only an explicitly negative response from the allowlist endpoint (the domain is present and the current host is not listed) causes a revocation.

PIN Gate on Source Changes

Changing an already-configured source URL (or switching from URL mode to USB mode) requires both a passing allowlist check and the kiosk PIN, verified via the Kiosk Plugin's /pin/verify endpoint. This is the same hashed, rate-limited PIN used for on-screen kiosk administration — there is no separate plaintext PIN for the PWA. First-ever setup on a fresh kiosk (where no source has been saved yet) is not PIN-gated, since there is nothing to protect yet.

5.4 Kiosk Fullscreen URL — Admin Toolbar Suppression

When the Kiosk Plugin's dedicated fullscreen URL is accessed, the WordPress admin toolbar is suppressed even for logged-in administrators. This is a deliberate security measure: a kiosk device in a public location should not display "Howdy, [admin name]" and a direct link into wp-admin to a curious visitor who wanders up to the screen. The shortcode embed ([lexipic_kiosk]) on a normal themed page retains the admin toolbar for logged-in users, since that is expected WordPress behaviour in a non-kiosk context.

5.5 LexiPic Plugin Uninstall

Deleting (not deactivating) the LexiPic Plugin triggers a complete, irreversible cleanup: database tables are dropped, all plugin options are deleted, all uploaded audio files are removed, all custom IME engine files are removed, and scheduled background events are cleared. This "clean uninstall" behaviour is intentional — a plugin that leaves orphaned data, options, and files behind is harder to audit. Export all data before deleting the plugin.

6. Deployment Scenarios

The three components of the suite can be combined in different ways to suit different institutional contexts.

6.1 Online Museum Kiosk (Full Suite)

A museum deploys a WordPress site with both the LexiPic Plugin (for community contribution) and the LexiPic Kiosk Plugin (for kiosk display). Each physical kiosk in the museum runs the Kiosk PWA in Remote URL mode, pointed at the museum's dedicated fullscreen kiosk URL. Curators add new word sets from any browser; changes appear on all kiosks automatically on the next sync.

- WordPress site: LexiPic Plugin + LexiPic Kiosk Plugin installed.
- Kiosk devices: Kiosk PWA in Remote URL mode, domain added to allowlist.
- Visitor experience: Fullscreen swipeable card archive, no browser chrome, idle-reset between visitors.

6.2 Offline / Travelling Exhibition (USB Mode)

A community organisation presents a travelling word archive at village events without reliable internet. Word sets are created and exported from a WordPress site running the LexiPic Plugin. The exported JSON files are placed on a USB drive in the lexipic-kiosk-data folder alongside a settings.json. The Kiosk PWA runs from the USB drive (or from a local copy on a laptop) in USB mode. No network connection is required at the event.

- WordPress site: LexiPic Plugin only (for authoring).
- Exhibition device: Kiosk PWA in USB mode, data folder on USB drive.
- Visitor experience: Same card-slider interface, fully offline.

6.3 Classroom / Library Embed

A school or library website already running WordPress adds the LexiPic Kiosk Plugin and embeds the kiosk on an existing page using [lexipic_kiosk]. No dedicated kiosk hardware is needed — any browser visiting the page sees the interactive word-card archive within the site's normal theme. Students can use the page on PCs, tablets, or mobiles.

- WordPress site: LexiPic Kiosk Plugin (LexiPic Plugin optional, for in-house authoring).
- No separate PWA required.
- Visitor experience: Inline kiosk inside the school or library website.

6.4 Standalone Archive (Single Plugin)

A researcher or community member wants a personal archive on their own WordPress site with no kiosk hardware. They install only the LexiPic Plugin and add [lexipic] to a page. Visitors browse, and authorised contributors add new entries, directly through the WordPress front end.

- WordPress site: LexiPic Plugin only.
- No Kiosk Plugin. No PWA.
- Visitor experience: Archive tab on a standard WordPress page.

7. Quick Reference

7.1 Shortcodes

Shortcode	Effect
[lexipic]	LexiPic Plugin — renders the full three-tab archive on any page.
[lexipic set="slug"]	LexiPic Plugin — renders the archive pre-loaded to a specific set.
[lexipic_kiosk]	Kiosk Plugin — renders the kiosk card player within the page.

7.2 Key URLs (Kiosk Plugin)

URL Pattern	Description
https://yoursite.com/?lexipic_kiosk=1	Fullscreen kiosk URL (default permalink).
https://yoursite.com/kiosk/	Fullscreen kiosk URL (pretty permalink).
https://yoursite.com/wp-json/lexipic-kiosk/v1/sets	REST: list all sets.
https://yoursite.com/wp-json/lexipic-kiosk/v1/allowlist	REST: domain allowlist (public).
https://yoursite.com/wp-json/lexipic/v1/sets	REST: LexiPic Plugin sets endpoint.

7.3 File and Directory Reference

Component	Path	Contents
LexiPic Plugin	wp-content/plugins/lexipic-plugin/	Plugin PHP, admin pages, front-end JS/CSS.
LexiPic Plugin	wp-content/uploads/lexipic/	Audio recordings, custom IME engine files.
Kiosk Plugin	wp-content/plugins/lexipic-kiosk/	Plugin PHP, admin pages, kiosk JS/CSS.
Kiosk Plugin	wp-content/uploads/lexipic-kiosk/	Extracted set images and audio files.
Kiosk PWA	lexipic-pwa-nt/index.html	PWA entry point.
Kiosk PWA	lexipic-pwa-nt/app.js	All PWA application logic.
Kiosk PWA	lexipic-pwa-nt/sw.js	Service worker for offline caching.

Kiosk PWA	lexipic-pwa-nt/lexipic-kiosk-data/	USB mode data folder (settings.json + set JSONs).

7.4 Languages Shipped with the LexiPic Plugin

Code	Language	Script
bpm	Bishnupriya Manipuri (ইমার ঠার)	Bengali script variant
asm	Assamese (অসমীয়া)	Assamese script
ben	Bengali (বাংলা)	Bengali script

7.5 Browser Support Notes

Feature	Browser Support
Web Speech API (voice capture)	Best in Chromium-based browsers (Chrome, Edge). Not supported in Firefox or Safari as of this release; field can be filled manually.
File System Access API (USB mode)	Chrome, Edge, and other Chromium browsers. Not available in Firefox or Safari; USB mode will not function.
IndexedDB (offline cache / USB handle)	All modern browsers, including Firefox and Safari.
Service Worker / PWA install	Chrome, Edge, Firefox, and Safari 16.4+.
Audio .webm playback	Chrome, Edge, Firefox. Not natively supported in Safari/iOS without FFmpeg .mp3 conversion.
Audio .mp3 playback	All modern browsers including Safari/iOS (requires FFmpeg on the server).

8. Troubleshooting

8.1 LexiPic Plugin

Audio recordings do not play in Safari or iOS

Recordings are stored as .webm files. Safari/iOS does not support .webm audio natively. Install FFmpeg on the server (most managed WordPress hosts support this on request) and the plugin will automatically convert new recordings to .mp3 in the background. The Settings page shows FFmpeg detection status. Existing .webm recordings need to be re-recorded after FFmpeg is enabled, as conversion is not applied retroactively.

Voice capture (Speak button) does not work

The Web Speech API is required. Ensure the browser is Chromium-based (Chrome or Edge) and that the site is served over HTTPS — the microphone API requires a secure context. On HTTP sites or unsupported browsers, the field remains manual-entry only.

IME produces incorrect script characters

Check that the correct language is selected for the set. If a custom forwardMap.json has been uploaded, review it for incorrect or conflicting mappings. Restore the bundled default from LexiPic → IME Engine → Manage Files if needed. The last 20 uploaded versions of each file are kept as backups.

8.2 Kiosk Plugin

The on-screen admin panel stays locked — PIN prompt appears but is rejected

Go to LexiPic Kiosk → Settings in wp-admin and confirm that a PIN has been set (the page shows whether one is configured). If the PIN has been forgotten, set a new one from wp-admin. There is no PIN recovery mechanism — the new PIN takes effect immediately.

Kiosk shows a blank page or loses content after a network interruption

The kiosk uses IndexedDB for local caching. If IndexedDB data was cleared (e.g. by the browser's storage management) or the kiosk has never fully loaded, the cache may be empty. Ensure the kiosk has network access when it first loads so it can populate the cache. Subsequent interruptions will be handled transparently.

Admin toolbar is visible on the fullscreen kiosk URL

Upgrade to Kiosk Plugin version 1.2.1 or later. Versions before 1.2.1 showed the admin toolbar to logged-in administrators on the fullscreen URL. Ensure no other plugin is re-adding the toolbar to this URL.

8.3 Kiosk PWA

"URL not in approved list" error when setting up Remote URL mode

The hostname of the URL you entered is not on the domain allowlist served by the ALLOWLIST_ENDPOINT configured in app.js. Ask the WordPress administrator of the target site to add your domain to LexiPic Kiosk → Approved Kiosk URLs. The allowlist check is an exact hostname match — include the correct subdomain (e.g. www.yoursite.com vs yoursite.com).

USB mode "Select folder" button does nothing or folder picker closes immediately

The File System Access API (showDirectoryPicker) is required for USB mode. This API is only available in Chromium-based browsers (Chrome, Edge). It is not available in Firefox or Safari. Ensure the PWA is opened in Chrome or Edge.

PWA does not install (no "Install" prompt in the browser)

Installation requires HTTPS, a valid Web App Manifest (manifest.json), and an active service worker (sw.js). If serving from a USB drive or local filesystem using the file:// protocol, installation is not available — serve the PWA from an HTTP server even locally (e.g. using Python's http.server or a simple localhost server). Service workers require either localhost or HTTPS.

Icons show as text (e.g. "folder_open" instead of an icon glyph)

The Material Icons font file (material-icons.woff2) must be present in the vendor/material-icons/ folder alongside the PWA files. If the folder was not included in deployment, re-copy it from the original lexipic-pwa-nt package. The font is intentionally vendored (not CDN-loaded) so icons work offline.

9. Changelog

LexiPic Plugin

Version 1.2.2 (Current)

- Current stable release.

LexiPic Kiosk Plugin

Version 1.2.1 (Current)

- Security fix: the dedicated fullscreen kiosk URL no longer shows the WordPress admin toolbar to logged-in administrators. Previously, a visitor at the kiosk could see the admin's name and a direct link into wp-admin if an admin had ever logged in using that browser.

Version 1.2.0

- New: "Approved Kiosk URLs" admin page and public /allowlist REST endpoint. Enables vetting of which domains the standalone Kiosk PWA is permitted to connect to in Remote URL mode.
- The PWA reuses the Kiosk Plugin's existing PIN (via /pin/verify) to gate source changes — no second PIN to manage.

Version 1.1.1

- Fix: clicking anywhere on a word card played its audio. Now only the Play button triggers audio playback.

Version 1.1.0

- Performance: word-card images and audio are now stored as files under wp-content/uploads/lexipic-kiosk/ instead of inline base64 in the database. Set lists now use a single batched database query.
- Existing sets are converted automatically on upgrade — no manual action required.
- The JSON export/import format is unchanged; older backups containing base64 media still import correctly.

Version 1.0.0

- Initial release.

Kiosk PWA

Version 2.1.0 (Current)

- Current stable release. Includes domain allowlist re-validation, PIN-gated source changes, IndexedDB handle persistence for USB mode, and service worker offline caching.

10. Appendix

A. Minimum Requirements Summary

Component	Requirement	Value
LexiPic Plugin	WordPress version	5.8 or later
LexiPic Plugin	PHP version	7.4 or later
LexiPic Plugin	FFmpeg (optional)	For .mp3 audio conversion
LexiPic Kiosk Plugin	WordPress version	6.0 or later
LexiPic Kiosk Plugin	PHP version	7.4 or later
Kiosk PWA	Browser (URL mode)	Any modern browser with HTTPS
Kiosk PWA	Browser (USB mode)	Chromium-based (Chrome/Edge) only
Kiosk PWA	Server requirement	None — static files only

B. Licencing

All three components of the LexiPic Language and Heritage Archive suite are released under the GNU General Public License version 2 or later (GPLv2+).

Component	Licence
LexiPic Plugin v1.2.2	GPLv2 or later
LexiPic Kiosk Plugin v1.2.1	GPLv2 or later
Kiosk PWA v2.1.0	GPLv2 or later
Material Icons (vendored)	Apache 2.0
Fraunces font (CDN)	SIL Open Font Licence 1.1

Full licence text: https://www.gnu.org/licenses/gpl-2.0.html

C. Links

Resource	URL
Developer / Author	https://bhoomitech.org

LexiPic Plugin documentation	https://bhoomitechapps.github.io/documentation/lexipic
LexiPic Kiosk Plugin documentation	https://bhoomitechapps.github.io/documentation/lp-k/
GNU GPLv2 Licence	https://www.gnu.org/licenses/gpl-2.0.html